

# Can Data Beat the Market?

## Predicting NBA Games and Evaluating Kalshi Odds

Exploratory Data Analysis Report

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GitHub repository: <https://github.com/USERNAME/kalshi-nba-eda>

**Key finding: Across 915 matched games, Kalshi's pregame home-team probabilities were broadly calibrated (3.2 percentage-point expected calibration error), with 68.5% directional accuracy and a 0.201 Brier score.**

### Summary of Research Questions

**1. Calibration.** Do final pregame Kalshi NBA prices correspond to actual win frequencies? **2. Model comparison.** Can leakage-safe NBA features support a model that beats Kalshi on accuracy, Brier score, and log loss? **3. Conditions.** When are market probabilities most and least accurate, especially across volume, probability, rest, and recent-form groups? **4. Simulated returns.** Do model-market gaps of 5%, 10%, or 15% yield positive historical returns after transaction costs?

### Motivation

Prediction-market prices summarize traders' information as tradable probabilities, but liquidity, public attention, and uncertainty may produce systematic errors. NBA games offer frequent outcomes and measurable pregame context. This EDA asks whether the market is calibrated and whether recent performance, rest, team strength, and volume expose useful structure before formal modeling. The objective is evaluation, not a claim that historical patterns guarantee profit.

Scope update: this report completes data collection, cleaning, feature construction, and exploratory analysis. Model tuning and simulated trading remain later-project tasks.

## Dataset

Two public sources were collected for the 2025-26 NBA season. Basketball Reference provided 1,322 completed schedule/result rows. Kalshi's live and historical APIs provided 2,720 settled team-contract records in the season window. Team names, dates, and opponents linked 1,123 games to one home-team contract; 915 of those games had a usable one-minute quote at least 15 minutes before the official scheduled tipoff. The final analytical table has 915 rows and 60 columns. Each row is one completed game; columns cover identifiers, teams, scores, Kalshi quotes and volume, outcomes, and leakage-safe rolling features.

| Stage                       | Rows  | Meaning                                      |
|-----------------------------|-------|----------------------------------------------|
| <b>Basketball Reference</b> | 1,322 | Completed 2025-26 games                      |
| <b>Kalshi contracts</b>     | 2,720 | One binary contract per team                 |
| <b>Matched games</b>        | 1,123 | Unique home-contract matches                 |
| <b>Analysis dataset</b>     | 915   | Valid pregame probability and settled result |

## Missing Data

The raw pipeline intentionally retains nonmatches: 199 schedule games (15.1%) lacked a unique Kalshi home-contract match, and 208 matched games (18.5%) lacked a valid quote inside the final 24 pregame hours. These rows are documented in the collection audit and excluded from calibration metrics rather than imputed. In the 915-row analytical table, probability, outcome, 24-hour pregame volume, spread, and Elo difference have 0% missingness. Ten games (1.1%) have missing rolling-form and rest differences because one or both teams had no earlier recorded game. They remain missing for EDA and will receive training-only neutral imputation or be excluded during modeling.

## Method

The Python pipeline paginated both Kalshi tiers, combined nine Basketball Reference monthly tables, normalized team labels, and matched games using home team, opponent, and an eight-hour tipoff-to-expiration window. It selected the last one-minute bid/ask midpoint ending at least 15 minutes before scheduled tipoff and summed only the final 24 hours of pregame candle volume. Games were sorted chronologically before computing 5- and 10-game win rates, point differentials, rest, back-to-back indicators, and Elo. No current-game score or future result enters a pregame feature.

## Variables of Interest

**kalshi\_prob** is the home contract's bid/ask midpoint and addresses calibration, model comparison, and trading thresholds. **home\_win** is the binary target. **pregame\_volume** and **bid\_ask\_spread** represent liquidity. **recent\_win\_pct\_5\_diff**, **recent\_point\_diff\_5\_diff**, **rest\_diff**, and **elo\_diff** represent pregame performance and schedule context for later models.

## EDA Results: Variable Summaries

**Kalshi probability** (n=915): mean 0.554; SD 0.227; min 0.035; Q1 0.385; median 0.570; Q3 0.745; max 0.985. **Bid/ask spread** (n=915): mean 0.012; SD 0.033; min 0.010; Q1 0.010; median 0.010; Q3 0.010; max 1.000.

**Final-24-hour pregame volume** (n=915): mean 448,615; SD 690,222; min 460; Q1 149,770; median 262,884; Q3 498,854; max 8,445,673.

**Recent point-differential difference (last 5)** (n=905): mean -0.143; SD 13.565; min -39.000; Q1 -9.400; median 0.000; Q3 8.600; max 40.600. **Rest-day difference** (n=905): mean 0.027; SD 0.959; min -8.000; Q1 0.000; median 0.000; Q3 0.000; max 7.000.

**Elo difference** (n=915): mean 2.387; SD 116.186; min -386.531; Q1 -71.723; median 0.000; Q3 71.428; max 407.623. **Per-game Brier score** (n=915): mean 0.201; SD 0.188; min 0.000; Q1 0.048; median 0.141; Q3 0.308; max 0.801.

**Categorical summaries.** home\_win: home=511, away=404. price\_source: bid/ask midpoint=915. Probability-bin counts from 0-10% through 90-100% were 19, 56, 60, 117, 133, 113, 133, 129, 124, and 31. Volume quartile counts were 229, 229, 228, and 229.

| Games | Accuracy | Brier score | Log loss | Calibration error |
|-------|----------|-------------|----------|-------------------|
| 915   | 68.5%    | 0.201       | 0.585    | 0.032             |

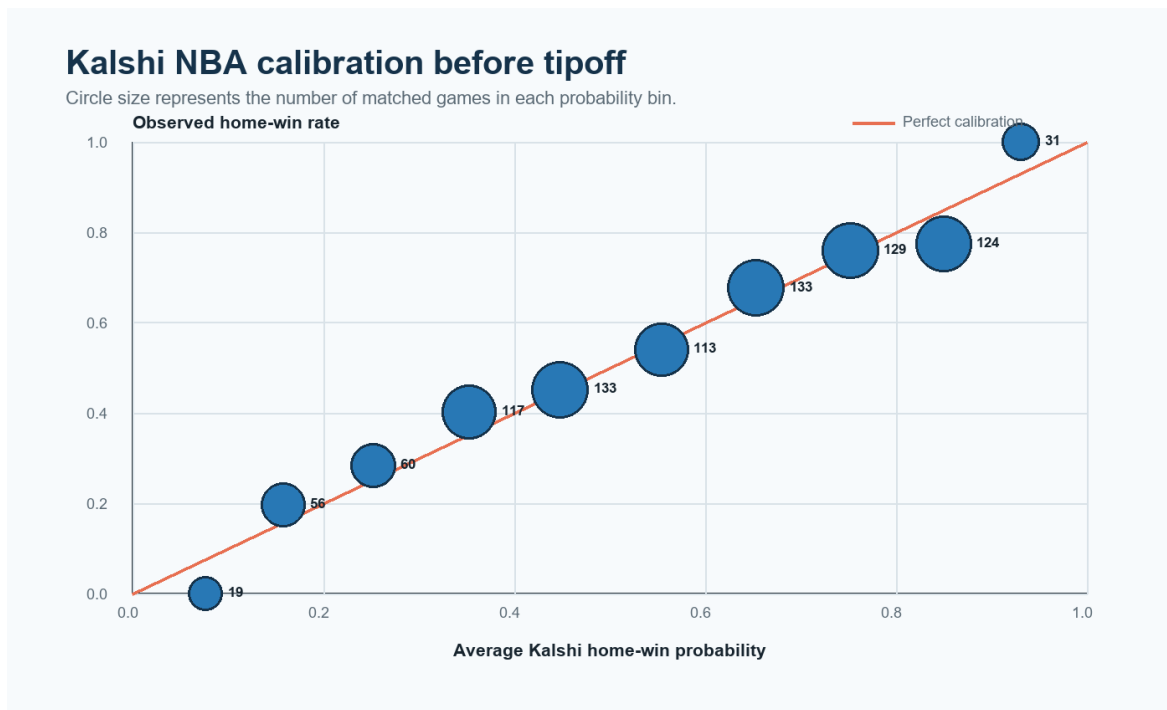


Figure 1. Calibration of Kalshi's final pregame home-team probabilities. Points near the diagonal indicate that average quoted probability and realized win rate agree.

The market is broadly calibrated: mean probability was 55.4% versus a 55.8% observed home-win rate. The largest well-populated deviation is the 80-90% bin, where an 85.0% average quote produced a 77.4% win rate. Extreme bins contain fewer games, so their apparent gaps are less stable. Calibration alone does not prove market efficiency, but it establishes a strong baseline for the proposed models.

## EDA Results: Conditions, Relationships, and Anomalies

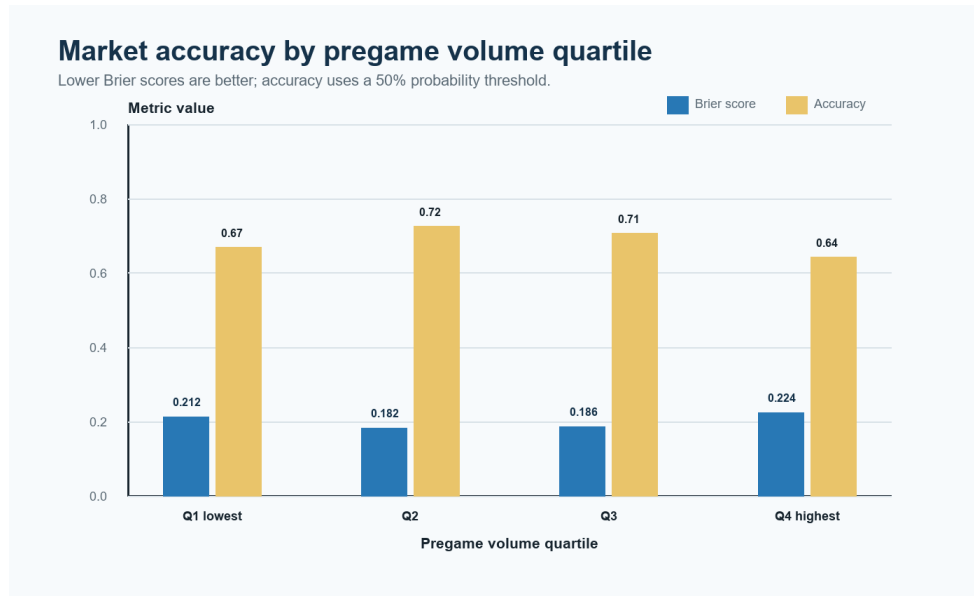


Figure 2. Accuracy and Brier score by final-24-hour pregame volume quartile. The middle quartiles outperform both tails, so the relationship is not monotonic.

Q2 had the lowest Brier score (0.182) and highest directional accuracy (72.5%). The highest-volume quartile performed worst (Brier 0.224; accuracy 64.2%), contrary to the simple expectation that more volume always means better pricing. This is descriptive, not causal: nationally prominent, playoff, or unusually uncertain games may attract both high volume and difficult-to-price outcomes.

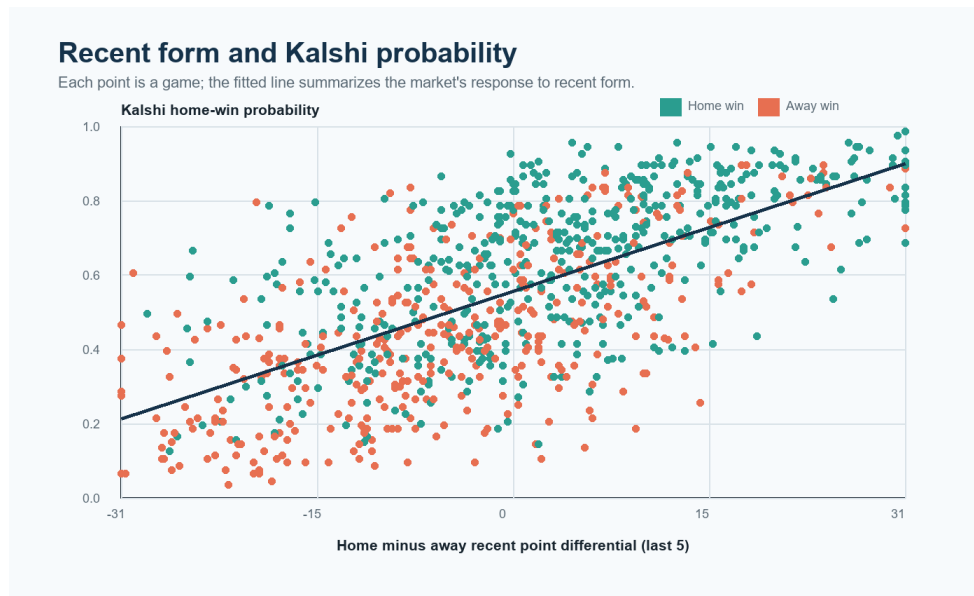


Figure 3. Kalshi probability versus the home-minus-away five-game point-differential difference. The upward fit shows that prices incorporate recent team form.

The positive relationship supports using recent form as a model feature, while the wide vertical spread shows that it is not sufficient by itself. The largest errors were upsets: three home teams priced at 89.5% lost. These cases deserve later review for injury news, lineup changes, and schedule context, but they should not be treated as a repeatable trading rule without out-of-sample validation.

## Challenge Goals Update

**Messy Data:** completed. The project now handles two Kalshi storage tiers, cursor pagination, nested JSON, per-market candles, monthly HTML tables, timezone conversion, team normalization, incomplete quotes, and audited matching. **Machine Learning:** unchanged but not yet executed in this EDA. Logistic regression, random forest, and gradient boosting will use the constructed features with a 60/20/20 chronological split. Hyperparameters will be chosen on validation data; the final period will be opened once for model-versus-market comparison.

## Work Plan Evaluation

**Download and inspect (3 h proposed):** longer than expected because of API partitioning and per-market candles. **Clean and merge (5 h):** substantially complete; matching and leakage audits were essential. **Pregame features (5 h):** complete and model-ready. **Modeling (6 h):** not started; 6-8 h remain. **Visualizations and writing (5 h):** EDA complete; final-model updates should require 3-4 h.

## Testing

The runnable test program asserts team-name mappings, unique game and ticker keys, probability bounds, the 15-minute cutoff, agreement between Kalshi settlement and NBA scores, and exact reproduction of the reported Brier score. A two-game synthetic table verifies that rolling features use only prior games. During development, audits caught and corrected two realistic failures: using expected market expiration instead of scheduled tipoff, and allowing an adjacent-date ticker to match twice. Visual review then checked calibration-bin counts, legends, axes, captions, and outliers.

## Collaboration

This project was completed individually. Resources consulted were Kalshi's official API documentation, Basketball Reference's 2025-26 schedule pages, and documentation for pandas, NumPy, Pillow, and ReportLab. No other people were consulted beyond course staff and the resources listed here.

## Limitations and Next Steps

The analytical sample excludes unmatched games and games without a recent pregame quote, so coverage may not be random. Volume measures only the final 24 pregame hours. EDA comparisons are descriptive and do not adjust for confounders. Next, the project will freeze the chronological split, tune models on validation data, evaluate the untouched test set, and simulate threshold trades with Kalshi transaction costs.

## References

Kalshi API: [https://docs.kalshi.com/getting\\_started/historical\\_data](https://docs.kalshi.com/getting_started/historical_data) and <https://docs.kalshi.com/api-reference/historical/get-historical-market-candlesticks>. Basketball Reference: [https://www.basketball-reference.com/leagues/NBA\\_2026\\_games.html](https://www.basketball-reference.com/leagues/NBA_2026_games.html). Original project proposal: [Evaluating\\_Kalshi\\_Odds.pdf](#).